# WEATHER API INTEGRATION PROJECT

## Internship Task 2 :

### 1. Title

Weather API Integration using Python

### 2. Objective

The objective of this project is to fetch real-time weather data from an online API using Python and display the results properly.
This project demonstrates API integration, JSON parsing, and data handling using the Requests module.

### 3. Problem Statement

Weather information is useful for daily activities and planning.
This project fetches weather details for a user-entered city and displays important information such as temperature, weather condition, and humidity.

### 4. Features Implemented

- Fetch weather data using API
- Use Requests module
- Parse JSON response
- User search functionality
- Display formatted weather information
- Exception handling

## 5.Create project file

- Create new file
- `weather_app.py`

## 6.Source code

weather_app.py

```python
import requests
city = input("Enter city: ")
url = f"https://wttr.in/{city}?format=j1"
try:
    response = requests.get(url)
    data = response.json()
    temperature = data["current_condition"][0]["temp_C"]
    weather = data["current_condition"][0]["weatherDesc"][0]["value"
    humidity = data["current_condition"][0]["humidity"]
    print("\nWeather Report")
    print("Temperature:", temperature, "°C")
    print("Condition:", weather)
    print("Humidity:", humidity)
except Exception as e:
    print("Error:", e)
```

## 7.Sample Output

```
PS C:\Users\Sanjana\Desktop\Task2_API integration> python weather.
Enter city: Hyderabad

------ WEATHER REPORT ------
City: Hyderabad
Temperature: 37 °C
Weather Condition: Sunny
Humidity: 35 %
PS C:\Users\Sanjana\Desktop\Task2_API integration>
```

## 8. Explanation of the Code

### Importing Requests Module

The Requests module is imported to send HTTP requests to the weather API.

### User Input

The user enters a city name to search weather information.

### API Request

The program sends a request to the wttr.in weather API.

**JSON Parsing**

The JSON response is converted into a Python dictionary using:

response.json()

**Displaying Results**

The program displays:

- Temperature
- Weather condition
- Humidity

**Exception Handling**

The try-except block prevents the program from crashing if errors occur.

**9. Conclusion**

This project successfully demonstrates API integration using Python. The program fetches real-time weather data, parses JSON responses, and displays useful weather information in a structured format.